

MySQL Assignment 2 – Querying Data

Clauses and Joins

1. Distinct Values:

- ❖ a query to retrieve distinct salaries from the Employees table.

Ans:

```
133 • select distinct salary from employees;
```

134

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
salary			
60000			
80000			
45000			
70000			
55000			
65000			
90000			
95000			
75000			
85000			
100000			

2. Alias (AS):

- ❖ Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

Ans:

```
134 • select age as Employee_Age, salary as Employee_Salary from employees;
```

135

136

Employee_Age	Employee_Salary
27	60000
31	80000
26	45000
29	70000
25	55000
28	65000
32	90000
27	70000
30	55000
29	80000
28	60000

3. Where Clause & Operators:

- ❖ Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

Ans:

```
135 • select * from employees
136     where salary >50000 and hire_date< '2016-01-01';
```

<									
Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:									
	employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
▶	5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000
	5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000
	5004	Kiara Malhotra	F	29	2015-07-05	NULL	8	3	70000
	5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1	55000
	5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

❖ Find the employee whose designation is missing and fill it with "Data Scientist".

Ans:

```
140 • update employees
141     SET designation = 'Data Scientist'
142     where designation is NULL AND employee_id=5004;
```

<									
Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:									
	employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
▶	5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000
	5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000
	5004	Kiara Malhotra	F	29	2015-07-05	Data Scientist	8	3	70000
	5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1	55000
	5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Sorting and Grouping Data:

1. ORDER BY:

❖ Find employees sorted by department ID in ascending order and salary in descending order.

Ans:

```
144 • select * from employees
145     order by department_id asc, salary desc;
```

<									
Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:									
	employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
▶	5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000
	5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	2	3	90000
	5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000
	5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	2	1	60000
	5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	3	4	100000
	5026	Vivaan Singh	M	29	2019-03-10	Business Analyst	3	2	75000
	5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000
	5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	4	4	100000
	5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4	60000
	5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000
	5024	Aryan Shetty	M	27	2018-11-05	Product Manager	5	1	95000

employees 6 ×

2. LIMIT:

- ❖ Display the first 5 employees hired in the year 2018.

Ans:

```
147 • select * from employees
148     where year(hire_date)=2018
149     order by hire_date asc
150     limit 5;
```

	employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
▶	5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000
	5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000
	5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	7	3	85000
	5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4	60000
	5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Aggregate Functions:

- ❖ Calculate the sum of all salaries in the Finance department.

Ans:

	Total_Fianace_Salary
▶	170000

- ❖ Find the minimum age among all employees.

Ans:

	minimum_age
▶	25

4. GROUP BY:

- ❖ List the maximum salary for each location.

Ans:

	location	max_salary
▶	Bangalore	95000
	Chennai	95000
	Hyderabad	90000
	Pune	100000

- ❖ Calculate the average salary for each designation containing the word 'Analyst'.

Ans:

Result Grid			Filter Rows:
	designation	avg_salary	
▶	Data Analyst	60000.0000	
	Marketing Analyst	60000.0000	
	Supply Chain Analyst	60000.0000	
	Financial Analyst	85000.0000	
	Finance Analyst	85000.0000	
	Business Analyst	75000.0000	
	Quality Assurance Analyst	80000.0000	

5. HAVING:

- ❖ Find departments with less than 3 employees.

Ans:

Result Grid			Filter Rows:
	department_name	employee_count	
▶	Business Development	2	
	Content Creation	2	
	Finance	2	
	Operations	1	
	Product Management	2	
	Research and Development	1	
	Software Development	1	

- ❖ Find locations with female employees whose average age is below 30.

Ans:

Result Grid			Filter
	location	avg_age	
▶	Bangalore	27.3333	
	Chennai	26.7500	
	Pune	29.2000	

Joins:

1. Inner Join:

- ❖ List employee names, their designations, and department names where employees are assigned to a department.

Ans:

Result Grid   Filter Rows: Export:  Wrap Cell Content: 			
	employee_name	designation	department_name
▶	Anvi Chaudhary	Business Development Executive	Business Development
	Kiaan Desai	Sales Executive	Business Development
	Ananya Paul	Content Writer	Content Creation
	Anaya Kapoor	Event Coordinator	Content Creation
	Aaradhya Iyer	Customer Support Executive	Customer Support
	Arnav Rao	Customer Success Manager	Customer Support
	Vihaan Mohan	Supply Chain Analyst	Customer Support
	Vihaan Singh	Data Analyst	Data Science
	Avni Iyengar	Data Scientist	Data Science
	Vivaan Singh	Business Analyst	Data Science
	Kiara Malhotra	Data Scientist	Design
	Dhruv Shetty	UI Developer	Design
	Diya Jha	Graphic Designer	Design
	Ishaan Kumar	Financial Analyst	Finance
	Advik Kapoor	Finance Analyst	Finance
	Zoya Khan	Legal Counsel	Human Resources
	Ishika Patel	Talent Acquisition Specialist	Human Resources
	Aadhya Iyengar	HR Specialist	Human Resources

2. Left Join:

- ❖ List all departments along with the total number of employees in each department, including departments with no employees.

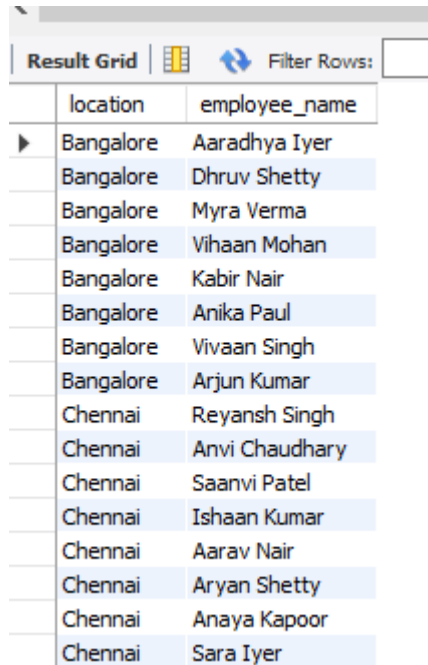
Ans:

Result Grid   Filter Rows:		
	department_name	Total_employees
▶	Business Development	2
	Content Creation	2
	Customer Support	3
	Data Science	3
	Design	3
	Finance	2
	Human Resources	3
	IT	4
	Marketing	3
	Operations	1
	Product Management	2
	Research and Develo...	1
	Software Development	1

3. Right Join:

- ❖ Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

Ans:



	location	employee_name
▶	Bangalore	Aaradhya Iyer
	Bangalore	Dhruv Shetty
	Bangalore	Myra Verma
	Bangalore	Vihaan Mohan
	Bangalore	Kabir Nair
	Bangalore	Anika Paul
	Bangalore	Vivaan Singh
	Bangalore	Arjun Kumar
	Chennai	Reyansh Singh
	Chennai	Anvi Chaudhary
	Chennai	Saanvi Patel
	Chennai	Ishaan Kumar
	Chennai	Aarav Nair
	Chennai	Aryan Shetty
	Chennai	Anaya Kapoor
	Chennai	Sara Iyer

Thank You...